

EDS Lab Assignments

Practical 5

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5.1.1 Stacked Plot

Problem Statement:

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Code:

```
import matplotlib.pyplot as plt
import pandas as pd
```

```
# Data for Months and Temperature for three cities
```

```
data = {
    'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August',
'September', 'October', 'November', 'December'],
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
}
```

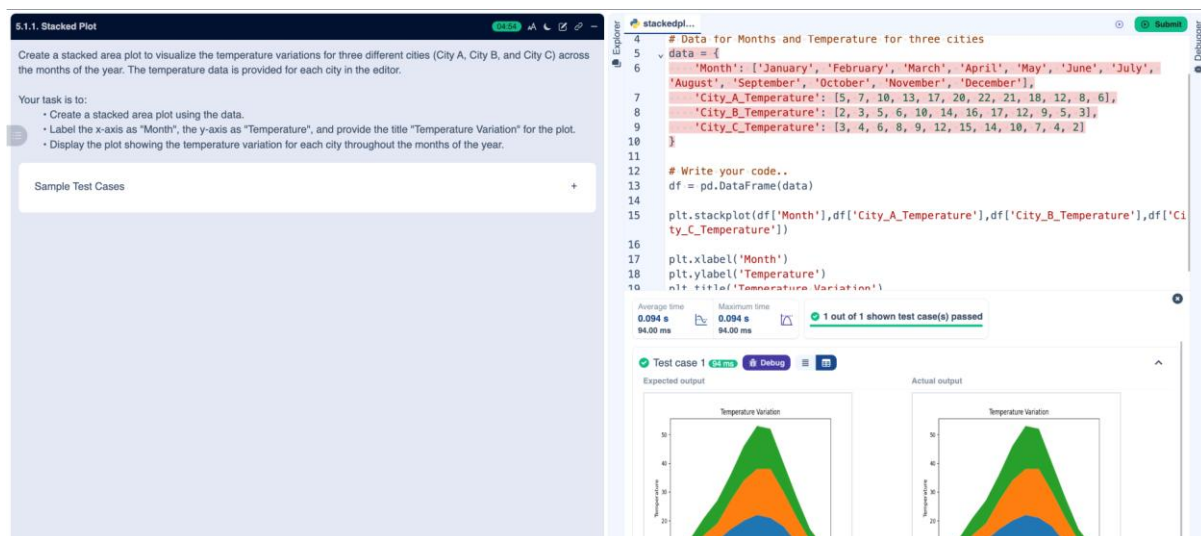
```
# Write your code... df =
pd.DataFrame(data)
```

```
plt.stackplot(df['Month'], df['City_A_Temperature'], df['City_B_Temperature'],
df['City_C_Temperature']) plt.xlabel('Month') plt.ylabel('Temperature')
```

```
plt.title('Temperature Variation')
```

```
plt.legend(loc = 'upper left')
```

```
plt.show() Output:
```



5.2.1 Titanic Dataset

Problem Statement:

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using `Matplotlib`.

The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (`Pclass` Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Pie Chart (`Gender` Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as **"Gender Distribution"**.

Histogram (`Age` Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use `lightgreen` for the bars with black edges (`edgecolor = 'black'`).
- Set the number of bins to **8** for the histogram.
- Title the plot as **"Age Distribution"**.
- Label the x-axis as **"Age"** and the y-axis as **"Frequency"**.

Bar Plot (`Survival` Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the `Survived` column.
- Use the colors `lightblue` for survivors (1) and `lightcoral` for non-survivors (0).
- Title the plot as **"Survival Count"**.
- Label the x-axis as **"Survived (0 = No, 1 = Yes)"** and the y-axis as **"Count"**.

Scatter Plot (`Fare` vs `Age`):

- Create a scatter plot to visualize the relationship between the Fare and Age of passengers.
- Use orange for the data points.
- Title the plot as **"Fare vs Age"**.
- Label the x-axis as **"Age"** and the y-axis as **"Fare"**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')
```

```
# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))
```

```
# Plot 1: Count of passengers by class
axes[0, 0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(),
               color='skyblue')
axes[0, 0].set_title("Passenger Class Distribution")
axes[0, 0].set_xlabel("Pclass")
axes[0, 0].set_ylabel("Count")
```

```
# Plot 2: Gender distribution
axes[0, 1].pie(df['Gender'].value_counts(),
               labels=df['Gender'].value_counts().index, autopct='%1.1f%%',
               colors=['lightblue', 'lightcoral'])
axes[0, 1].set_title("Gender Distribution")
```

```
# Plot 3: Age distribution
axes[1, 0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')
axes[1, 0].set_title("Age Distribution")
axes[1, 0].set_xlabel("Age")
axes[1, 0].set_ylabel("Frequency")
```

```
# Plot 4: Survival count
axes[1, 1].bar(df['Survived'].value_counts().index, df['Survived'].value_counts(),
               color=['lightblue', 'lightcoral'])
axes[1, 1].set_title("Survival Count")
axes[1, 1].set_xlabel("Survived (0 = No, 1 = Yes)")
axes[1, 1].set_ylabel("Count")
```

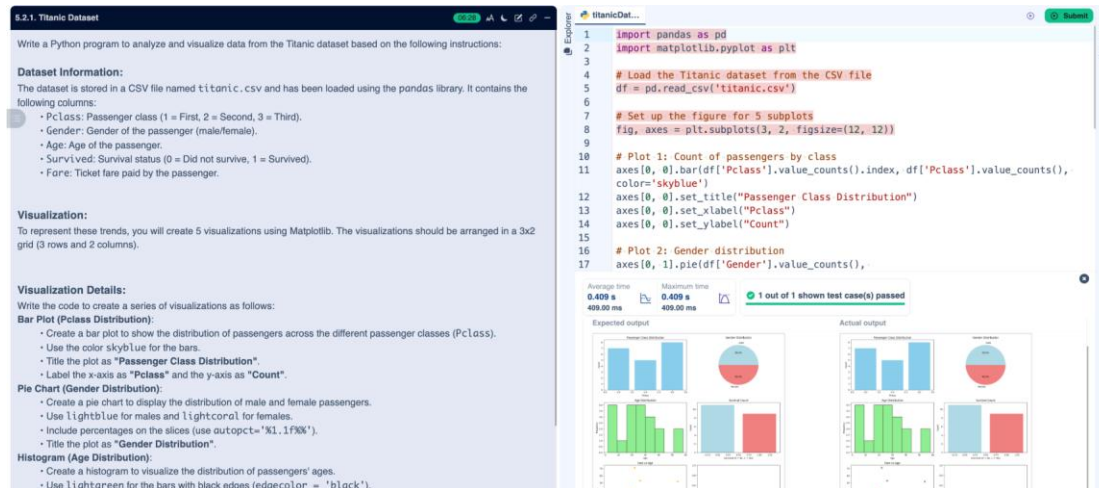
```
# Plot 5: Fare vs Age
axes[2, 0].scatter(df['Age'], df['Fare'], color='orange', edgecolors='black')
axes[2, 0].set_title("Fare vs Age")
axes[2, 0].set_xlabel("Age")
```

```
axes[2, 0].set_ylabel("Fare")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:



5.2.2 Histogram of Passenger Information of Titanic

Problem Statement:

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Histogram

plt.hist(data['Age'], bins = 30, edgecolor = 'k')

plt.title("Age Distribution")
plt.xlabel('Age')

plt.ylabel("Frequency")
plt.show()
```

Output:

4.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use 30 bins for the histogram.
2. Set the edge color of the bars to black (k).
3. Label the x-axis as 'Age' and the y-axis as 'Frequency'.
4. Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cunha, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,275350,0.0,S
6,0,3,"Moran, Mr. James",male,0,0,330877,0.4583,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.6625,E66,S
8,0,3,"Peltson, Master. Guste Leonard",male,2,1,1,349089,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achee)",female,14,1,0,237736,30.0706,C
```

Note: Refer to the visible test case for better reference.

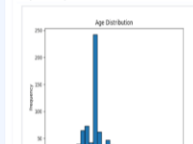
Sample Test Cases

Histogram...

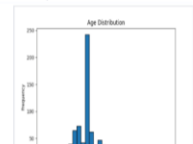
```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 plt.hist(data['Age'], bins=30, edgecolor='k')
17
18 plt.title('Age Distribution')
19
20 plt.xlabel('Age')
21 plt.ylabel('Frequency')
```

Average time: 0.111 s
Maximum time: 0.111 s
1 out of 1 shown test case(s) passed

Expected output



Actual output



5.2.3 Bar Plot of Survival Rate of Passengers

Problem Statement:

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the '**Survived**' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count**" to the chart.
4. Label the x-axis as '**Survived**' and the y-axis as '**Count**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True) data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)
```

```
# Write your code here for Bar Plot for Survival Rate
```

```
survival_counts = data['Survived'].value_counts()
```

```
survival_counts.plot(kind='bar') plt.title('Survival
Count') plt.xlabel('Survived') plt.ylabel('Count')
```

```
plt.show() Output:
```


5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).

2. Set the chart type to 'bar'.

3. Add the title 'Survival Count' to the chart.

4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	21.0	1	0	15170	7.25	S	
2	1	1	Cooking, Mrs. John Bradley (Florence Briggs Thayer)	female	38.1	0	0	17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	517062	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35.0	0	0	174508	8.85	S	
6	0	3	Woods, Mr. James	male	0.0	0	0	338637	8.4583	U	
7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	S	
8	0	3	Pallasson, Master. Gosta Leonard	male	2.3	1	0	349989	21.075	S	
9	1	3	Jonsson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	0	347742	11.1333	S	
10	1	2	Asser, Mrs. Nicholas (Adele Achen)	female	34.1	0	0	237736	30.8788	C	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,21.0,1,0,15170,7.25,S

2,1,1,"Cooking, Mrs. John Bradley (Florence Briggs Thayer)",female,38.1,0,0,17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26.0,0,0,517062,7.925,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35.0,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35.0,0,0,174508,8.85,S

6,0,3,"Woods, Mr. James",male,0.0,0,338637,8.4583,U

7,0,1,"McCarthy, Mr. Timothy J",male,54.0,0,0,17463,51.8625,S

8,0,3,"Pallasson, Master. Gosta Leonard",male,2.3,1,349989,21.075,S

9,1,3,"Jonsson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27.0,0,0,347742,11.1333,S

10,1,2,"Asser, Mrs. Nicholas (Adele Achen)",female,34.1,0,237736,30.8788,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf

import pandas as pd

import matplotlib.pyplot as plt

Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

Write your code here for Bar Plot for Survival Rate

survival_counts = data['Survived'].value_counts()

survival_counts.plot(kind='bar')

plt.title('Survival Count')

plt.xlabel('Survived')

plt.ylabel('Count')

plt.show()

Average time

0.130 s

Maximum time

0.130 s

100.00 ms

100.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Survival Count

Bar chart showing survival counts: Survived=0 has a count of approximately 170, and Survived=1 has a count of approximately 230.

Survival Count

Bar chart showing survival counts: Survived=0 has a count of approximately 170, and Survived=1 has a count of approximately 230.

5.2.4 Bar Plot for Survival by Gender

Problem Statement:

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Gender"** to the chart.
4. Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
5. The legend should indicate **'Not Survived'** and **'Survived'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True) data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Gender
survived_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
survived_counts.plot(kind = 'bar',stacked = True)

plt.title('Survival by Gender')
plt.xlabel('Gender') plt.ylabel('Count')
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output:

3.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "Survival by Gender" to the chart.
4. Label the x-axis as 'Gender' and the y-axis as 'Count'.
5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,517,7.25,,S
2,1,1,"Cassidy, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Meekinen, Miss. Laina",female,26,0,0,51706/02, 3181282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373459,8.45,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,0.4583,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.6625,646,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Masser, Mrs. Nicholas (Adele Aches)",female,34,1,0,237736,30.4708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

```
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

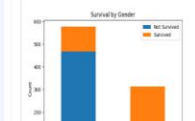
# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

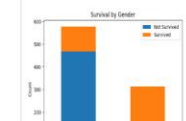
# Write your code here for Bar Plot for Survival by Gender
survival_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
survival_counts.plot(kind='bar', stacked=True)
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Average time: 0.146 s
Maximum time: 0.146 s
1 out of 1 shown test case(s) passed

Expected output



Actual output



5.2.5 Bar Plot for Survival by Pclass

Problem Statement:

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "**Survival by Pclass**" to the chart.
4. Label the x-axis as '**Pclass**' and the y-axis as '**Count**'.
5. The legend should indicate '**Not Survived**' and '**Survived**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True) data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)
```

```
# Write your code here for Bar Plot for Survival by Pclass
survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)
survival_counts.plot(kind = 'bar', stacked=True)
```

```
plt.title('Survival by Pclass')
plt.xlabel('Pclass')
plt.ylabel('Count')
```

```
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output:

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using `value_counts()`.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "Survival by Pclass" to the chart.
4. Label the x-axis as "Pclass" and the y-axis as "Count".
5. The legend should indicate "Not Survived" and "Survived".

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brandy, Mr. Owen Harris	male	22	1	0	A/5 21371,7	25	5	
2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71	2833	C85,C	
3	1	3	Hesketh, Miss. Laina	female	26	0	0	STON/O2	3181282	7	Q25,S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113883	53	1	C123,S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8	45,S	
6	0	3	Moran, Mr. James	male	0	0	0	338637	8	4583	0
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51	8625	E46,S
8	0	3	Palsson, Master. Gosta Leonard	male	2	1	1	349989	21	475,S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11	1333,S	
10	1	2	Nasser, Mrs. Nicholas (Adele Achee)	female	14	1	0	237736	38	0780,C	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Brandy, Mr. Owen Harris",male,22,1,0,A/5 21371,7,25,,5
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71,2833,C85,C
3,1,3,"Hesketh, Miss. Laina",female,26,0,0,STON/O2 3181282,7,925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113883,53,1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8,45,,S
6,0,3,"Moran, Mr. James",male,0,0,338637,8,4583,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51,8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349989,21,475,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11,1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achee)",female,14,1,0,237736,38,0780,,C
```

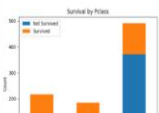
Note:

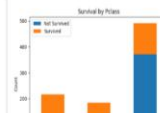
- Refer to the visible test case for better reference.
- Ensure you use the `groupby()` function with `value_counts()` to count the survivors and non-survivors for each Pclass.
- Do not manually use `size()` or `unstack()` without `value_counts()`. Use the `value_counts()` method for counting survival status directly.

BarPlotOf...

```
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Pclass
17 survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)
18
19 survival_counts.plot(kind='bar', stacked=True)
20 plt.title('Survival by Pclass')
21 plt.xlabel('Pclass')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

Average time: 0.145 s, Maximum time: 0.145 s, 1 out of 1 shown test case(s) passed

Expected output: 

Actual output: 

5.2.6 Bar Plot for Survival by Embarked

Problem Statement:

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked** " to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True) data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Embarked
survival_counts = data.groupby('Embarked_Q')
['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind = 'bar', stacked = True)
plt.title('Survival by Embarked')
plt.xlabel('Embarked') plt.ylabel('Count')
plt.legend(['Not Survived', 'Survived']) plt.show()
```

Output:

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

- Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
- Set the chart type to 'bar' and make it stacked.
- Add the title "Survival by Embarked" to the chart.
- Label the x-axis as 'Embarked' and the y-axis as 'Count'.
- Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
3,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,3,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,CBS,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2, 3181282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113883,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.85,S
6,0,3,"Moran, Mr. James",male,0,0,338877,0,4583,0
7,0,3,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349989,21.875,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,9.6786,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Embarked
survival_counts = data.groupby('Embarked_Q')[
    ['Survived'].value_counts().unstack().fillna(0)
]
survival_counts.plot(kind='bar', stacked=True)
plt.title('Survival by Embarked')
plt.xlabel('Embarked')
plt.ylabel('Count')
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Average time: 0.146 s
Maximum time: 0.146 s
146.20 ms
146.00 ms
1 out of 1 shown test case(s) passed

Expected output

Actual output

5.2.7 Box Plot for Age Distribution

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True) data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Pclass

data.boxplot(column = 'Age', by = 'Pclass')

plt.title('Age by Pclass') plt.suptitle("")

plt.xlabel('Pclass') plt.ylabel('Age')

plt.show()
```

Output:

3.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

- Use the Pclass column to group the data for the boxplot.
- Set the title of the plot to 'Age by Pclass'.
- Remove the default subtitle with plt.suptitle('').
- Label the x-axis as 'Pclass' and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,517,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2, 3101262,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.45,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,0.000,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.6625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Aches)",female,14,1,0,237736,30.0700,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotf...

```
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

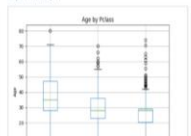
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Pclass
data.boxplot(column='Age', by='Pclass')
plt.title('Age by Pclass')
plt.suptitle('')
plt.xlabel('Pclass')
plt.ylabel('Age')
plt.show()
```

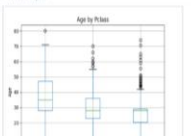
1 out of 1 shown test case(s) passed

Test Case 1

Expected output



Actual output



Problem Statement:

5.2.8 Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Survived

plt.figure(figsize=(8,6))
data.boxplot(column='Age', by='Survived')
plt.title('Age by Survival')
plt.suptitle('')
plt.xlabel('Survived')
plt.ylabel('Age')
plt.show()
```

Output:

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:
1. Use the Survived column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to "Age by Survival".
3. Remove the default subtitle with plt.suptitle().
4. Label the x-axis as "Survived" and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cooking, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Meikinen, Miss. Laina	female	26	0	0	STON/O2	33091282	7.925	S
4	1	1	Putrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	375369	8.45	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	0	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Polsson, Master. Gosta Leonard	male	2	3	1	349909	21.875	S	
9	1	1	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347462	11.1333	S	
10	1	2	Nasser, Mrs. Nicholas (Adele Achen)	female	14	1	0	237736	30.0708	C	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cooking, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Meikinen, Miss. Laina",female,26,0,0,STON/O2 33091282,7.925,S
4,1,1,"Putrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,375369,8.45,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Polsson, Master. Gosta Leonard",male,2,3,1,349909,21.875,S
9,1,1,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347462,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlot...

import matplotlib.pyplot as plt
Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
Write your code here for Box Plot for Age by Survived
plt.figure(figsize=(8,6))
data.boxplot(column='Age',by='Survived')
plt.title('Age by Survival')
plt.suptitle('')
plt.xlabel('Survived')
plt.ylabel('Age')
plt.show()

Average time: 0.126 s, Maximum time: 0.136 s, 1 out of 1 shown test case(s) passed
Expected output: Actual output:

Problem Statement:

5.2.9 Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass

plt.figure(figsize=(8,6))
data.boxplot(column='Fare', by='Pclass')
plt.title('Fare by Pclass')
plt.suptitle('')
plt.xlabel('Pclass')
plt.ylabel('Fare')
plt.show()
```

Output:

1.2.6 Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the Pclass column to group the data for the boxplot.

2. Set the title of the plot to "Fare by Pclass".

3. Remove the default subtitle with plt.suptitle('').

4. Label the x-axis as 'Pclass' and the y-axis as 'Fare'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101262,5.52,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.45,,S
6,0,3,"Moran, Mr. James",male,0,0,338077,0.00,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.0025,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0700,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotF...

3

4

5

6

7

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23

24

Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

Write your code here for Box Plot for Fare by Pclass

plt.figure(figsize=(8,6))

data.boxplot(column='Fare', by = 'Pclass')

plt.title('Fare by Pclass')

plt.suptitle('')

plt.xlabel('Pclass')

plt.ylabel('Fare')

plt.show()

Average time

0.126 s

126.00 ms

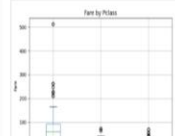
Maximum time

0.126 s

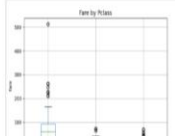
126.00 ms

1 out of 1 shown test case(s) passed

Fare by Pclass



Fare by Pclass



Problem Statement:

5.2.10 Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True) data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass

plt.figure() plt.scatter(data['Age'],
data['Fare']) plt.title('Age vs. Fare')
plt.xlabel('Age') plt.ylabel('Fare')
plt.show()
```

Output:

5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Set the title of the plot to **"Age vs. Fare"**.

3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	"Braund, Mr. Owen Harris"	male	21	1	0	4	5393	1	7.25
2	1	1	"Cumings, Mrs. John Bradley (Florence Briggs Thayer)"	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	"Heikkinen, Miss. Laina"	female	26	0	0	STON/O2	51.0		S
4	1	1	"Futrelle, Mrs. Jacques Heath (Lily May Peel)"	female	35	1	0	113803	53.1	C123	S
5	0	3	"Allen, Mr. William Henry"	male	35	0	0	17460	8.05		S
6	0	3	"Moran, Mr. James"	male	0	0	0	338637	5.40		Q
7	0	1	"McCarthy, Mr. Timothy J"	male	54	0	0	17463	51.0		S
8	0	3	"Palsson, Master. Gosta Leonard"	male	2	3	1	349909	21.075		S
9	1	3	"Jansson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)"	female	27	0	2	347742	11.1333		S
10	1	2	"Nasser, Mrs. Nicholas (Adele Achen)"	female	34	1	0	237736	30.07		C

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,21,1,0,4,5393,1,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2,51.0,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,17460,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,338637,5.40,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.0,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,S
9,1,3,"Jansson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,34,1,0,237736,30.07,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFareB...

3

Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

6

Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)

11

Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

Write your code here for Box Plot for Fare by Pclass

17

plt.figure()

plt.scatter(data['Age'], data['Fare'])

plt.title('Age vs. Fare')

plt.xlabel('Age')

plt.ylabel('Fare')

plt.show()

Average time

0.110 s

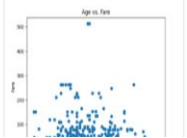
Maximum time

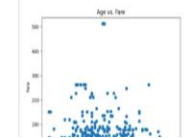
0.110 s

1 out of 1 shown test case(s) passed

Expected output

Actual output





Problem Statement:

5.2.11 Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status.

The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True) data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True) # Write your code here for
Scatter Plot for Age vs. Fare by Survived

colors = data['Survived'].map({0:'red', 1: 'blue'})
plt.scatter(data['Age'], data['Fare'], c=colors)

plt.title("Age vs. Fare by Survival")
plt.xlabel('Age') plt.ylabel('Fare')
plt.show()
```

Output:

3.2.11: Scatter Plot for Age vs. Fare by Survival

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Color the points based on the **Survived** column: Red for passengers who did not survive (**Survived = 0**). Blue for passengers who survived (**Survived = 1**).

3. Set the title of the plot to **"Age vs. Fare by Survival"**.

4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cooking, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17596,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,174608,0.85,,5
6,0,3,"Moran, Mr. James",male,0,0,330877,0.4583,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.6625,E46,S
8,0,3,"Pelisson, Master. Guste Leonard",male,2,3,1,349989,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,5
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFare...

Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

Write your code here for Scatter Plot for Age vs. Fare by Survival
colors = data['Survived'].map({0: 'red', 1: 'blue'})
plt.scatter(data['Age'], data['Fare'], c=colors)

plt.title("Age vs. Fare by Survival")
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()

Average time
0.123 s
123.00 ms

Maximum time
0.123 s
123.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output